

Project Work Plan

Course Name: Software Development I

Course Code: CSE 2216

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Project Title: Bounce Ball

Course: Software Development Lab

Programming Language: Java (Swing)

10-Week Work Plan for Bounce Ball Project

Week	Phase
1	Project Planning & Concept Finalization
2	Requirements Analysis & System Design
3	Environment Setup & Project Initialization
4	UI Layout & Game Window Development
5	Ball Movement & Paddle Implementation
6	Core Logic: Collision & State
7	Multiple Levels Implementation
8	Score System & Game Over Screen
9	Error Handling & Testing
10	Final Testing, Documentation & Submission

Week 1: Project Planning & Concept Finalization

- Select project topic and objectives.
- Define ball physics, obstacle mechanics, and project scope.

Prepare initial project plan.

Week 2: Requirements Analysis & System Design

- Gather functional requirements for gameplay, levels, and scoring.
- Design system architecture and class structure (OOP design).
- Create basic wireframes for the game window and HUD.

Week 3: Environment Setup & Project Initialization

- Install JDK and configure the IDE.
- Configure the Java Swing environment.
- Create project structure and main game window class.

Week 4: UI Layout & Game Window Development

- Develop the main game window and play area panel.
- Create the score display and status area.
- Apply theme and basic styling to the GUI.

Week 5: Ball Movement & Paddle Implementation

- Implement ball movement using the Timer class for the game loop.
- Add paddle/player control and keyboard input handling.
- Apply boundary collision so the ball bounces off walls.

Week 6: Core Logic: Collision & State

- Implement obstacle collision detection logic.
- Handle ball-paddle and ball-obstacle interactions.
- Manage overall game state (running, paused, game over).

Week 7: Multiple Levels Implementation

- Design and implement multiple levels with varying obstacle layouts.
- Add level progression and difficulty scaling.
- Implement level transition screens.

Week 8: Score System & Game Over Screen

- Implement the score system based on obstacles cleared and time.
- Add the game-over screen with final score display.
- Improve user experience and visual feedback.

Week 9: Error Handling & Testing

- Add exception handling for invalid states and edge cases.
- Test ball physics and collision detection across all levels.
- Fix bugs and optimize game performance.

Week 10: Final Testing, Documentation & Submission

- Perform complete system testing across all levels.
- Prepare project documentation and screenshots.
- Finalize and submit the project.